

Name _____ Date _____ Period _____

Solving Absolute Value Linear Equations - Practice

Algebra II | Unit 2 | Lesson 5 Practice | TEK A2.6.E

Objective: I can solve absolute value linear equations by isolating the absolute value and splitting into two cases.

Section 1 — Recall

1. Solve $|x| = 14$.

$x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

2. Solve $|x - 5| = 9$.

$x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Section 2 — Apply

3. Solve $|2x + 1| - 3 = 8$.

$x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

4. Solve $3|x - 4| = 21$.

$x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Section 3 — Challenge

5. Solve $4|x + 2| - 5 = 15$.

$x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

6. Solve $2|3x - 1| + 4 = 22$.

$x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Section 4 — Explain It

7. Explain, in your own words, the full process for solving an absolute value equation - isolate, split, solve, and (mention) check. Use one of today's problems as your example.
